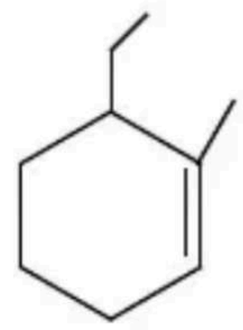
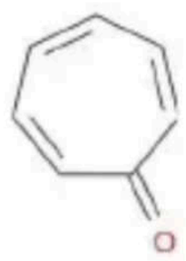
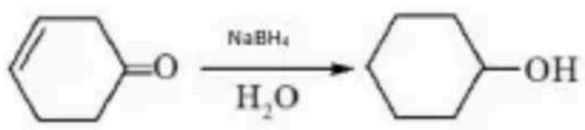
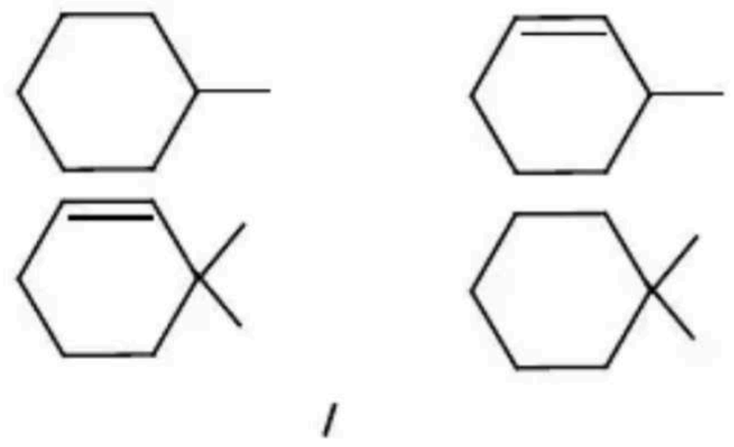
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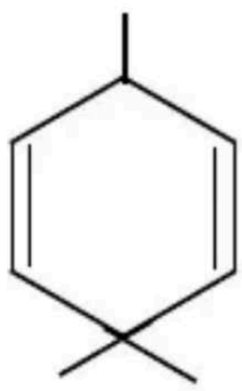

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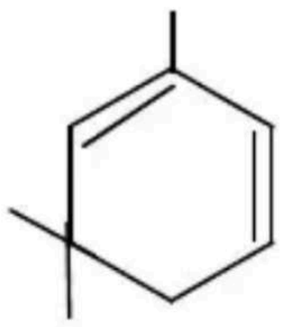

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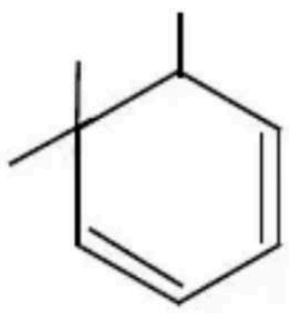


cis-2-butene







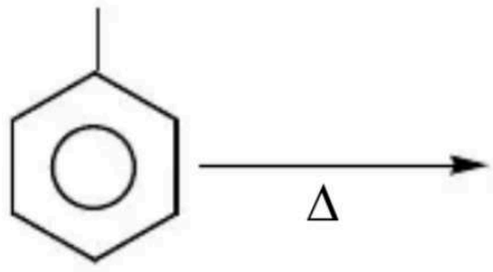


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$mol\ L^{-}$

$mol\ L^{-}$

$mol\ L^{-}$



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$O^{+} < O^{-} < O^{+}$

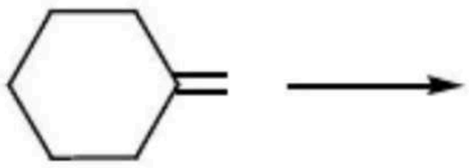
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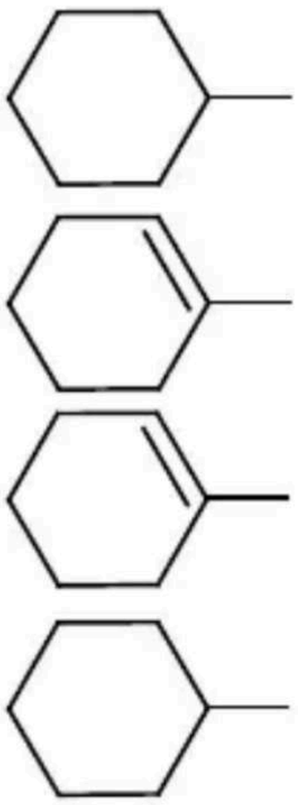
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$$\times \quad ^{-} \qquad\qquad \times \quad ^{-}$$

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$$GeX$$

$$SnF$$

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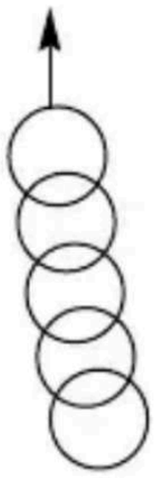
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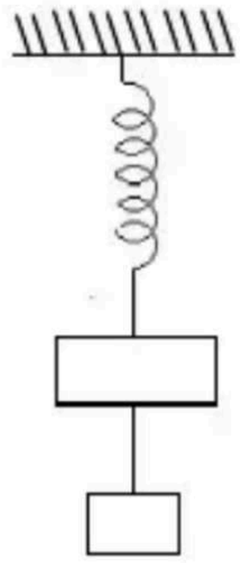
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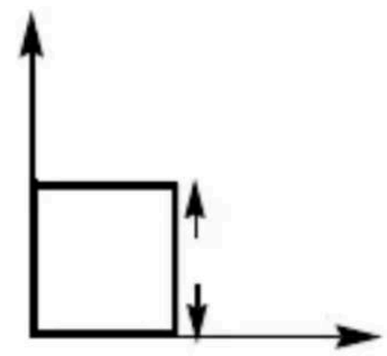
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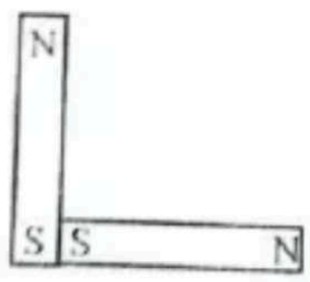
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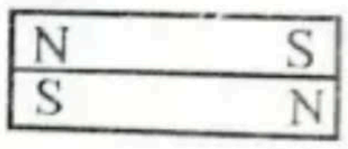
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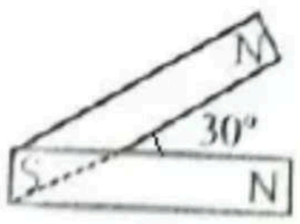
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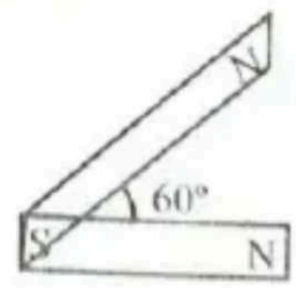
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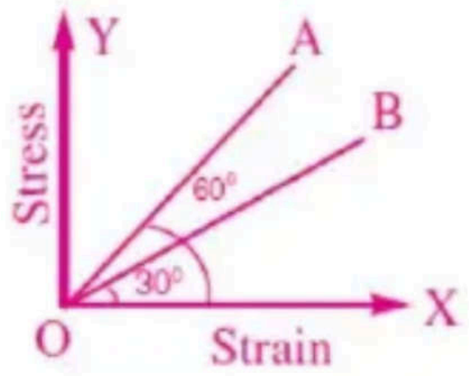
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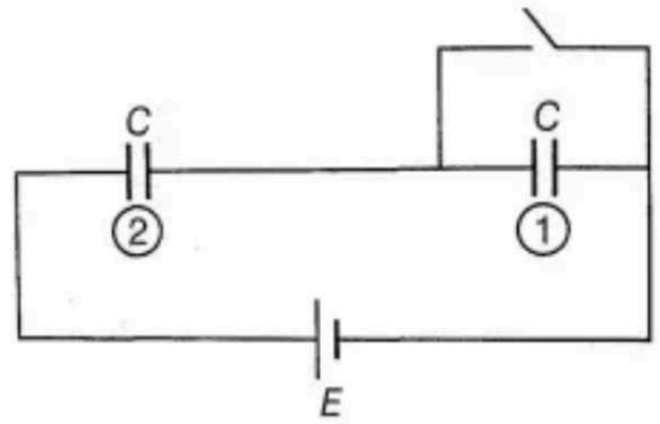
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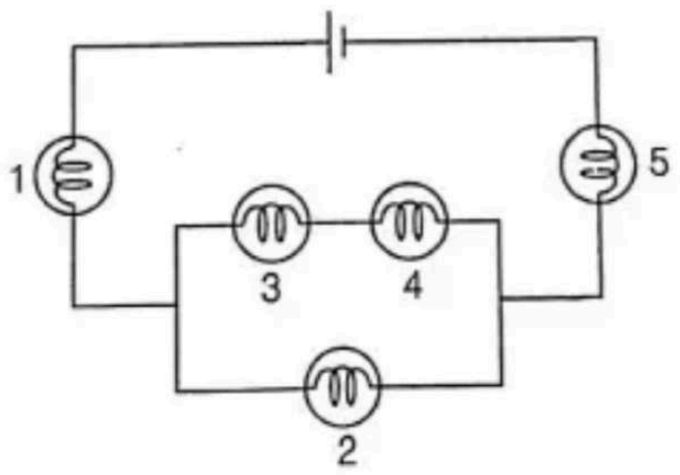


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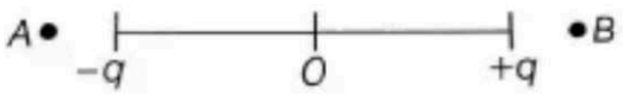
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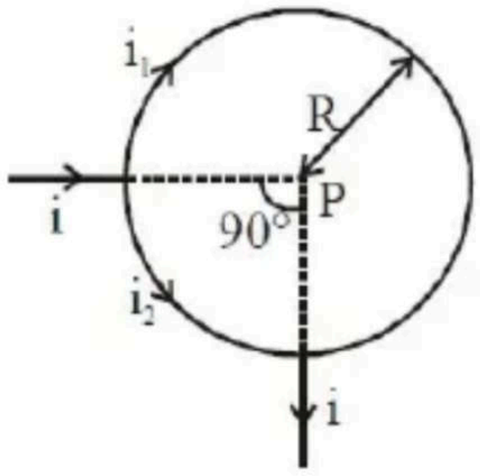
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$V_C = V_D > V_A = V_B$

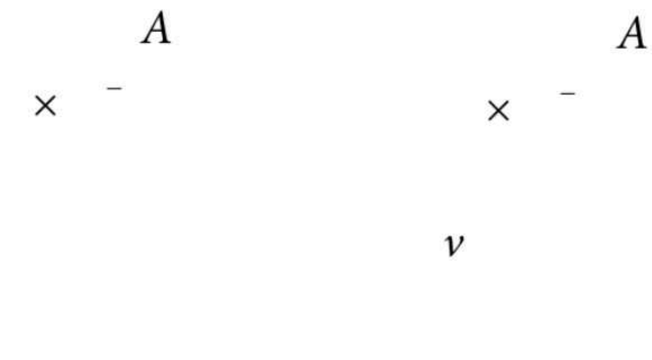
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$V_B > V_C = V_D > V_A$



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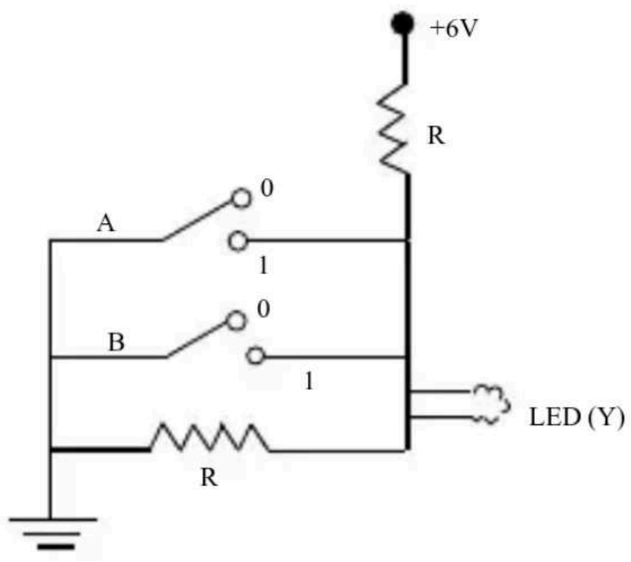
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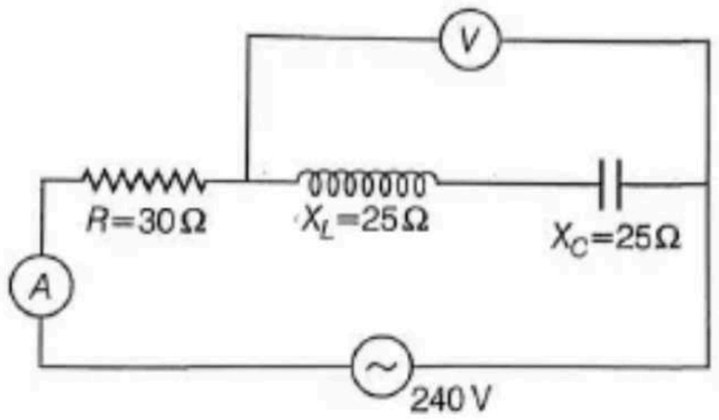
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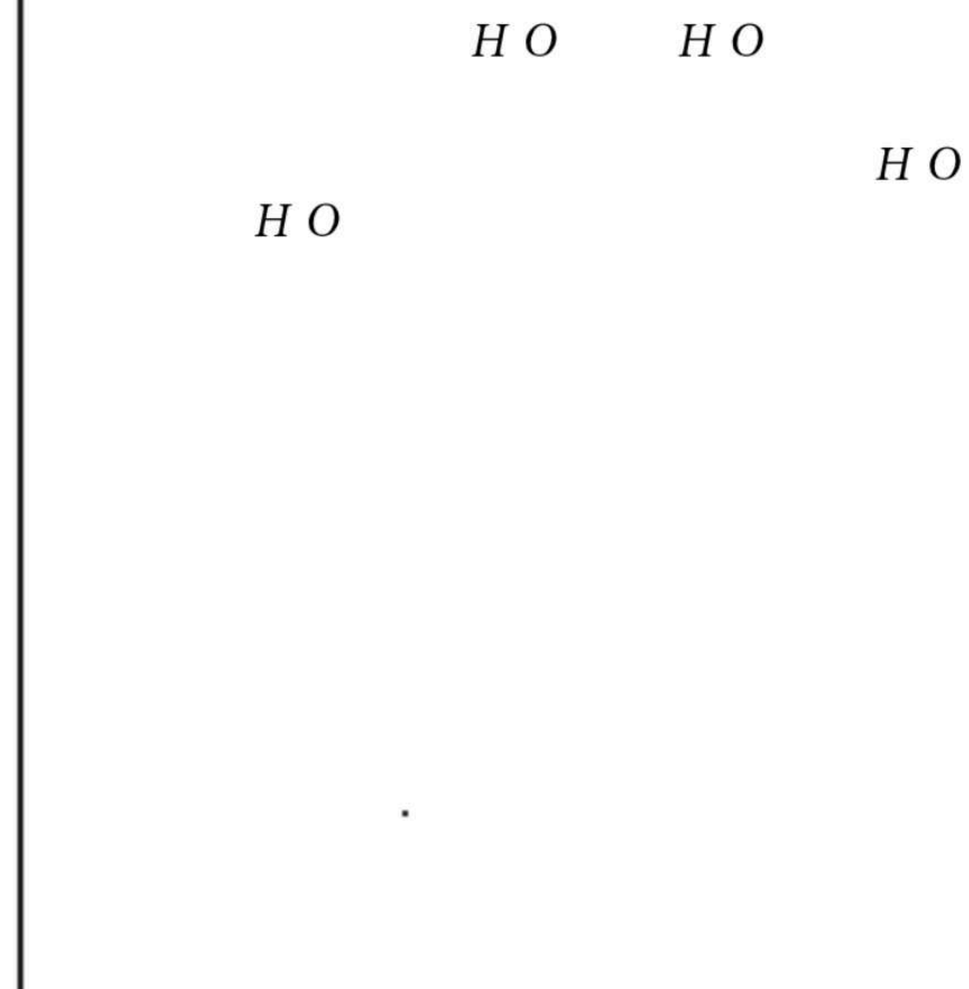
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$$+CH\ O+H\ O$$

F

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H O
H O
H O
NAD+

149. In which of the following phase proteins are synthesised in preparation for mitosis while cell growth continues ?

(1) G (2) S (3) G (4) G_o

150. How many are associated with G_o stage?

(I) Exit of G phase
(II) Metabolically active
(III) No longer proliferation
(IV) Suspended stage of cell cycle

(1) I & II only
(2) II,III & IV only
(3) I, II,& III only
(4) I, II, III,& IV

Section-A

151. Which of the following is correct regarding sexual dimorphism in animals?

(1) Male stingrays have pelvic claspers
(2) Male frogs have vocal sacs
(3) Male cockroaches have anal styles
(4) Male roundworms are longer than females

152. Given below are two statements.

Darwin called minor heritable variations as saltations.

Darwin believed that adaptive ability is due to the inheritance of acquired characters.

In light of the above statements, choose the correct answer from the options given below

(1) Both Statement I and Statement II are true
(2) Statement I is true but Statement II is false
(3) Statement I is false but Statement II is true
(4) Both Statement I and Statement II are false.

153. Identify the animal based on the characters given below.

(A) Forelimbs adapted for flying
(B) Four-chambered heart
(C) Body covered by hair
(D) Viviparity

The above characters are seen in

(1) *Macropus* (2) *Pteropus*
(3) *Corvus* (4) *Exocoetus*

154. Match List-I with List-II.

	List-I		List-II
(A)	Cerebral hemispheres	(I)	Pneumotaxic centre
(B)	Pons	(II)	Limbic system
(C)	Hippocampus	(III)	Corpus callosum
(D)	Hypothalamus	(IV)	Secretes neurohormones

(1) III I IV II
(2) III IV I II
(3) III I II IV
(4) III IV II I

155. Comb plates are found in

(1) *Chaetopleura*
(2) *Limulus*
(3) *Chelone*
(4) *Pleurobrachia*

Oxygen dissociation curve shifts to left at tissues.

High $H^+$and low pO_2at body tissues reduce the affinity of O_2 with haemoglobin.

(1) Both A and R are true but R is not the correct explanation of A.
(2) A is true but R is false
(3) Both A and R are true and R is the correct explanation of A.
(4) A is false but R is true

157. Given below are two statements.

Corpus luteum is maintained by the hormone LH.

Ovary secretes estrogen, progesterone and relaxin.

In light of the above statements, choose the correct answer from the options given below.

(1) Both Statement I and Statement II are true
(2) Statement I is true but Statement II is false
(3) Statement I is false but Statement II is true
(4) Both Statement I and Statement II are false.

158. Given below are two statements.

The second meiotic division in the secondary oocyte is completed after the fusion of male and female pronuclei.

Sex of human baby is determined by the sperm of father.

In light of the above statements, choose the correct answer from the options given below.

(1) Both Statement I and Statement II are true
(2) Statement I is true but Statement II is false
(3) Statement I is false but Statement II is true
(4) Both Statement I and Statement II are false.

159. Which of the following statements is true regarding haemoglobin in humans?

(1) It is an iron-containing protein.
(2) It transports CO_2 in the form of carbonic acid.
(3) It is 12-16 mg/100ml in a healthy human
(4) Its β chain synthesis is regulated by a gene on 16th chromosome.

160. Match List I with List II and choose the correct option.

(A)	Bronchioles	(I)	Ciliated epithelium
(B)	Epiglottis	(II)	Squamous epithelium
(C)	Alveoli	(III)	Striated muscle fibres
(D)	External intercostal muscles	(IV)	Elastic cartilage
(E)	Myometrium	(V)	Smooth muscle fibres

(1)	I	IV	II	III	V
(2)	I	IV	II	V	III
(3)	II	III	I	IV	V
(4)	I	IV	III	II	V

161. Which of the following is a narrow utilitarian argument for conserving biodiversity?

(1) 25000 species of plants contribute to traditional medicines
(2) Humans derive economic benefits through bioprospecting
(3) Watching spring flowers in full boom
(4) Agriculture and apiculture practices for food

162. The Bidder's canal of male frog

(1) is located in the testes
(2) carries urine and sperms
(3) directly opens into cloaca
(4) communicates with urinogenital duct

163. Given below are two statements.

Nodal tissue of the human heart is specialised cardiac musculature.

The number of action potentials generated per minute does not vary at different parts of nodal tissue.

In light of the above statements, choose the correct answer from the options given below.

(1) Both Statement I and Statement II are true
(2) Statement I is true but Statement II is false
(3) Statement I is false but Statement II is true
(4) Both Statement I and Statement II are false.

164. Which of the following is correctly matched?

(1) α-lactalbumin – Used in the treatment of emphysema
(2) HIV vaccine – Attenuated vaccine produced from yeast
(3) Humulin – Recombinant therapeutic drug for the treatment of diabetes mellitus
(4) PCR – Based on the principle of antigen-antibody interaction

165. Parturition is induced by a complex neuroendocrine mechanism involving cortisol, estrogens and oxytocin. Cortisol is secreted by the

(1) placenta
(2) ovary
(3) posterior pituitary
(4) adrenal gland

166. Which of the following is a correct combination?

(1)	Fibrous joint	Immovable	Cranial sutures
(2)	Hinge joint	Synovial fluid	Knee joint
(3)	Carti-laginous joint	Permits limited movements	Joints between adjacent vertebrae
(4)	Pivot joint	Allows movements	Joints be-tween skull and atlas

167. The majority of nephrons in the human kidney have
(1) short loops of Henle
(2) highly developed vasa recta
(3) long loops of Henle
(4) no peritubular capillary network

168. Choose the correct set of statements related to predation.
A. It is detrimental to both species.
B. It is a conduit for energy transfer in a food chain.
C. It maintains species diversity by reducing interspecific competition among prey species.
D. Herbivores, in a broad ecological context, are very different from predators.
E. Chemical methods adopted in pest control are based on predation.
(1) A, B only (2) D, E only
(3) B, C only (4) A, C only

169. Which hormone of the pituitary regulates the growth and secretion of milk in mammary glands?
(1) Prolactin
(2) Estrogen
(3) Progesterone
(4) Growth hormone

170. Which of the following statements holds true for sickle cell anaemia but not for thalassemia?
(1) It is an autosomal recessive disorder.
(2) It reduces the rate of synthesis of α or β chains of globin.
(3) It is due to a mutation of the HBB gene of the 16th chromosome.
(4) Qualitative problem of synthesis of β chain of globin.

Given below are two statements.

The Mediterranean orchid *Ophrys* employs 'sexual deceit' to get pollination done by a species of bee.

A given fig species can be pollinated only by its 'partner' wasp species and no other species.

In light of the above statements, choose the correct answer from the options given below.
(1) Statement I is false but Statement II is true
(2) Statement I is true but Statement II is false
(3) Both Statement I and Statement II are false.
(4) Both Statement I and Statement II are true

172. Match list I with List II and choose correct option.

(A)	Charas	(I)	Erythroxylum coca
(B)	Crack	(II)	*Papaver som-niferum*
(C)	Reserpine	(III)	*Cannabis sativa*
(D)	Smack	(IV)	*Raawolfia vomito-ria*

(1)	I	II	IV	III
(2)	III	IV	I	II
(3)	III	I	IV	II
(4)	III	II	I	IV

173. Which of the following has myosin binding site?
(1) Troponin (2) Dystrophin
(3) Actin (4) Tropomyosin

174. Which of the following assisted reproductive technologies does not involve *in vivo* fertilisation?
(1) ICSI (2) GIFT
(3) Artificial insemination
(4) Intrauterine insemination

175. Choose the correct set of placental mammals of Australia.
(1) Ant eater, numbat
(2) Lemur, flying squirrel
(3) Sugar glider, bandicoot
(4) Bobcat, Tasmanian tiger cat

176. Which of the following is about the scapula of the human skeleton?
(1) It is a dorsal, flat triangular bone in the thorax.
(2) Its spine projects as acromion process.
(3) Its glenoid cavity articulates with the head of the humerus.
(4) It is a part of the pelvic girdle.

177. Which of the following is an alien species in Indian rivers that poses a threat to native fishes?
(1) Nile perch
(2) *Parthenium*
(3) African catfish
(4) *Eicchornia*

178. Which of the following is a sexually transmitted infection that is curable if detected early?
(1) Hepatitis –B
(2) Genital herpes
(3) HIV infection
(4) Chlamydiasis

179. A woman who carries the gene for haemophilia can transmit the recessive gene to
(1) sons only
(2) daughters only
(3) both daughters and sons
(4) all sons and few daughters

180. Which of the following statements are about antibodies?
A. They are a type of proteins produced by T-lymphocytes.
B. They are involved in humoral immunity.
C. Thymosins stimulate their production.
D. IgA antibodies in colostrum provide passive immunity to the newborn.
E. They are cytokine barriers.
(1) A, E only
(2) B, D only
(3) C, D only
(4) B, C, D

181. Expiratory capacity is
(1) the total volume of air a person can expire after normal inspiration
(2) volume of air that remains in the lungs after a normal expiration
(3) maximum volume of air a person can expire after a forced inspiration
(4) volume of air in the lungs after forcible expiration

182. In a person, the volume of blood at the end of ventricular systole is 60 mL, and the volume of blood at the end of ventricular diastole is 10 mL. His heart rate is 80 beats per minute. Calculate the cardiac output.
(1) 4 litres
(2) 5 litres
(3) 3.5 litres
(4) 5600 ml

183. The efferent arteriole of human kidneys has lower levels of urea and glucose than the afferent arteriole.
Peritubular capillaries arise from efferent arteriole.
(1) Both A and R are true and R is the correct explanation of A
(2) Both A and R are true but R is not the correct explanation of A
(3) A is true but R is false
(4) A is false but R is true

184. The primary carnivores are
(1) primary consumers
(2) first trophic level organisms
(3) secondary consumers
(4) second trophic level organisms

185. Which of the following is an approach for the conservation of threatened animals in their natural habitat without human interference?
(1) Wildlife safari parks
(2) Zoological parks
(3) Cryopreservation techniques
(4) Wildlife sanctuaries

Section-B

186. Match the contents of List I with List II and choose the correct option.

	List I		List II
(A)	*Anopheles*	(I)	Radula
(B)	*Aplysia*	(II)	Notochord in larval tail
(C)	*Ascidia*	(III)	Compound eyes, antennae
(D)	*Asterias*	(IV)	Calcareous ossicles

(1) III II I IV
(2) III I II IV
(3) III IV I II
(4) III II IV I

187. Which of the following does occur in the fallopian tube?
(1) Completion of second meiotic division in primary oocyte
(2) Fusion of sperm and ovum
(3) Cleavage leading to the formation of morula
(4) Muscular movement for transport of gametes

188. Which of the following statements are true regarding the proximal convoluted tubule of the human nephron?
A. It is lined by columnar epithelium.
B. It is situated in the medullary region of the kidney.
C. It secretes H^+ ions into the filtrate.
D. Conditional reabsorption of Na^+ aided by aldosterone occurs in it.
E. 70 to 80% of water and electrolyte reabsorption occur in it.
(1) C, E
(2) A, C
(3) B, D
(4) A, E

189. Ringworm infection is caused by
(1) *Trichomonas*
(2) *Epidermophyton*
(3) *Mycobacterium*
(4) Psiolophyton

Glucagon is a hyperglycemic hormone.

It stimulates the process of glycogenolysis and gluconeogenesis.

(1) Both A and R are true but R is not the correct explanation of A.
(2) A is true but R is false
(3) Both A and R are true and R is the correct explanation of A.
(4) A is false but R is true.

191. Match List I with List II and choose the correct option.

	List I		List II
(A)	Rheumatoid arthritis	(I)	Mendelian disorder
(B)	Phenylketonuria	(II)	Chromosomal disorder
(C)	Addison's disease	(III)	Autoimmune disorder
(D)	Turner's syndrome	(IV)	Hormonal disorder

(1) III II I IV
(2) III I IV II
(3) III IV II I
(4) III IV I II

192. Expressed sequence tags are
(1) all coding and non-coding sequences
(2) genes that are expressed as RNA
(3) amino acid sequences of proteins
(4) repetitive DNA sequences

193. Which of the following is the correctly matched set regarding the partial pressure of O_2 and CO_2?
(1) Dorsal aorta – pO_2=95, pCO_2=45
(2) Pulmonary artery – pO_2=95, pCO_2=40
(3) Pulmonary vein – pO_2=75, pCO_2=45
(4) Systemic tissues – pO_2=40, pCO_2=45

194. Given below are two statements.

The death of organisms in a trophic level marks the beginning of the detritus food chain.

Dead biomass serves as an energy source for saprotrophs.

In light of the above statements, choose the correct answer from the options given below.

(1) Statement I is true but Statement II is false

(2) Statement I is false but Statement II is true

(3) Both Statement I and Statement II are true

(4) Both Statement I and Statement II are false.

195. Choose the statement.

(1) The cranial capacity of the first human-like being, the hominid, is 650-800cc.

(2) *Australopithecus* lived in Near East and Central Asia.

(3) Pre-historic cave art developed about 18000 years ago.

(4) Neanderthal man used hides to protect their body and buried their dead.

196. Glucocorticoids secreted by the adrenal cortex are involved in

A. gluconeogenesis

B. reabsorption of Na^+ and water

C. maintenance of cardiovascular system

D. inflammatory reactions

Which of the above are correct?

(1) A, B

(2) B, C

(3) A, C

(4) All are correct

197. Which of the following is an autoimmune disorder?

(1) Rheumatoid arthritis

(2) Grave's disease

(3) Myasthenia gravis

(4) Myotonic dystrophy

198.

A J-shaped population growth curve represents exponential growth.

The fitness of a species in terms of its 'r' is significantly higher in the presence of a competing species.

(1) Both A and R are true and R is the correct explanation of A.

(2) Both A and R are true but R is not the correct explanation of A.

(3) A is true but R is false

(4) A is false but R is true

199. The cerebral aqueduct is found in

(1) midbrain

(2) cerebrum

(3) cerebellum

(4) hypothalamus

200. Match List I with List II and choose the correct option.

(A)	Vaults	(I)	Prevent entry of sperms into cervix
(B)	LNG - 20	(II)	cervix hostile to sperms
(C)	Oral pills	(III)	Prevent insemination
(D)	Condoms	(IV)	Inhibit ovulation

(1)	III	II	IV	I
(2)	I	IV	II	III
(3)	I	II	III	IV
(4)	I	II	IV	III